Confirmatory Factor Analysis

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Introduction

Confirmatory factor analysis (CFA) tests a pre-specified factor structure against data: which items load on which factors, which loadings are constrained to zero, and how factors correlate with each other. It is the measurement-model portion of structural equation modelling and the standard tool for validating questionnaire structures, latent-trait models in psychometrics, and multi-indicator latent constructs in clinical and behavioural research. Unlike exploratory factor analysis, CFA does not estimate which items belong to which factor — that decision is in the model — and instead estimates only the strength of the predetermined relationships and tests model-data fit.

Prerequisites

A working understanding of exploratory factor analysis, regression, and the `lavaan` model-syntax conventions for specifying latent variables.

Theory

CFA specifies a complete pattern matrix in advance: each item either loads freely on its assigned factor or is constrained to zero on every other factor. Fit is assessed via:

- **Chi-squared test**: a non-significant χ^2 formally accepts the model, but the test rejects too easily at large n.
- **Incremental fit indices**: CFI and TLI compare to a baseline null model; values above 0.95 indicate good fit, 0.90–0.95 acceptable.
- **Absolute fit**: RMSEA measures discrepancy per degree of freedom; below 0.06 is good, 0.06–0.08 acceptable. The accompanying 90 % CI quantifies precision.
- **SRMR**: standardised root-mean-square residual; below 0.08 indicates good fit.

A well-fitting model satisfies multiple criteria simultaneously — no single index is sufficient.

Assumptions

A correctly specified factor structure, multivariate Normal continuous indicators (or robust / WLSMV estimators for non-Normal or ordinal data), and sufficient sample size (rule of thumb: at least 10 observations per estimated parameter, or $n \geq 200$).

R Implementation

```r
library(lavaan); library(semTools)

# Three-factor model on HolzingerSwineford1939 dataset
model <- '
  visual  =~ x1 + x2 + x3
  textual =~ x4 + x5 + x6
  speed   =~ x7 + x8 + x9
'
```

```r
fit <- cfa(model, data = HolzingerSwineford1939)
summary(fit, fit.measures = TRUE, standardized = TRUE)

# Modification indices for potential improvements
head(modificationIndices(fit, sort = TRUE), 10)
```

Output & Results

`summary(fit, fit.measures = TRUE, standardized = TRUE)` reports unstandardised and standardised loadings, factor correlations, residual variances, and the full panel of fit indices (chi-squared, CFI, TLI, RMSEA with CI, SRMR). `modificationIndices()` lists potential parameter additions ordered by expected chi-squared improvement.

Interpretation

A reporting sentence: "The three-factor CFA fit the data adequately ($\chi^2(24) = 85.3$, $p < 0.001$; CFI 0.93; RMSEA 0.092 [90 % CI 0.071 to 0.114]; SRMR 0.065); all standardised loadings exceeded 0.50 and were statistically significant ($p < 0.001$). RMSEA exceeded the conventional 0.06 threshold, suggesting modest local misspecification — modification indices indicated a residual correlation between x_7 and x_9 that would improve fit if substantively justified." Always report multiple fit indices; no single one is decisive.

Practical Tips

- Do not follow modification indices blindly; each unconstrained parameter moves the analysis from confirmatory to exploratory and inflates Type I error if not pre-specified.
- For ordinal Likert items, use diagonally-weighted least squares (`cfa(..., estimator = "DWLS")` or `"WLSMV"`); ordinary ML on Likert items violates Normality assumptions.
- For non-Normal continuous data, use robust ML (`estimator = "MLR"`); standard errors and chi-squared are corrected for non-Normality.
- Measurement-invariance testing across groups (configural, metric, scalar invariance) is a standard follow-up before comparing latent means; use `semTools::measurementInvariance()`.
- Report chi-squared with degrees of freedom and the full set of fit indices; never report a single index alone.
- Standardised loadings (factor-loading correlations) communicate effect size more clearly than unstandardised; always include them.

R Packages Used

`lavaan` for `cfa()` and `sem()` with multiple estimators; `semTools` for measurement-invariance testing, reliability coefficients, and modification heuristics; `lavaanPlot` and `semPlot` for path-diagram visualisation; `blavaan` for Bayesian CFA when small samples or complex priors require it.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA

- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE